

Q.No.2

For $X|f$

Given $V = \frac{X-32}{2}$, $\Sigma f = 400$, $\Sigma fV = -91$, $\Sigma fV^2 = 2435$
 $\Sigma fV^3 = -1915$, $\Sigma fV^4 = 32339$

Solⁿ

$$N = \Sigma f = 400$$

Moments about origin for variable V

$$\mu'_r(V) = \frac{1}{N} \Sigma fV^r, \quad \text{we get } \mu'_1(V) = \frac{1}{N} \Sigma fV = -0.2275$$

$$\mu'_2(V) = \frac{1}{N} \Sigma fV^2 = 6.0875$$

$$\mu'_3(V) = \frac{1}{N} \Sigma fV^3 = -4.7875$$

$$\mu'_4(V) = \frac{1}{N} \Sigma fV^4 = 80.8475$$

Central Moments of variable V

$$\mu_2(V) = \mu'_2(V) - \mu'_1(V)^2$$

$$\mu_3(V) = \mu'_3(V) - 3\mu'_2(V)\mu'_1(V) + 2\mu'_1(V)^3$$

$$\mu_4(V) = \mu'_4(V) - 4\mu'_3(V)\mu'_1(V) + 6\mu'_2(V)\mu'_1(V)^2 - 3\mu'_1(V)^4$$

using above formulae/expressions, we get

$$\mu_2(V) = 6.035744, \quad \mu_3(V) = -0.6563303, \quad \mu_4(V) = 78.37324$$

For variable X $\mu_r(X) = h^r \mu_r(V)$

$$\text{we get } \mu_2(X) = 24.14297$$

$$\mu_3(X) = -5.250643$$

$$\mu_4(X) = 1253.972$$

$$\beta_1(X) = \frac{\mu_3^2(X)}{\mu_2^3(X)} = 0.601959082$$

$$\beta_2(X) = \frac{\mu_4(X)}{\mu_2^2(X)} = 2.151326$$

$\beta_1(X) > 0$, Skewed, $\mu_3(X)$ is -ve, hence it is negatively skewed.

$\beta_2(X) < 3$ data is platykurtic (highly flat)

Q.No.1

Solution End Sem Exam, Prob. + Stat. (MCC505)

The coefficient of variation (C.V.) is defined as

$$C.V. = \frac{S.D.}{Mean} \quad \text{or} \quad C.V. = \frac{S.D.}{Mean} \times 100 \%$$

use of C.V. : To compare the variability or consistency of different sets of data when i) means are not same and ii) the units of measurements are not same.

C.V. is a unit free quantity. $(\frac{1}{2} + \frac{1}{2} = 1 \text{ mark})$

Solⁿs : In usual notations, we are given

$$n_1 = 100, \bar{x}_1 = 15, \sigma_1 = 3, \quad n = n_1 + n_2 = 250,$$

$$\bar{x} = 15.6, \quad \sigma = \sqrt{13.44} \Rightarrow \sigma^2 = 13.44$$

We want σ_2 ?

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \Rightarrow 15.6 = \frac{100 \times 15 + 150 \bar{x}_2}{250}$$

$$\Rightarrow 150 \bar{x}_2 = 250 \times 15.6 - 1500 = 2400$$

$$\text{or } \bar{x}_2 = \frac{2400}{150} = 16$$

$$\boxed{\bar{x}_2 = 16} \rightarrow (1 \text{ mark})$$

$$\text{Hence, } d_1 = \bar{x}_1 - \bar{x} = 15 - 15.6 = -0.6 \quad \text{and } d_2 = \bar{x}_2 - \bar{x} = 16 - 15.6 = 0.4$$

The variance of the combined group is given by

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)}$$

$$\Rightarrow 250 \times 13.44 = 100(9 + 0.36) + 150(\sigma_2^2 + 0.16)$$

$$\Rightarrow 150 \sigma_2^2 = 250 \times 13.44 - 100 \times 9.36 - 150 \times 0.16 = 2400$$

$$\Rightarrow \sigma_2^2 = \frac{2400}{150} = 16 \Rightarrow \boxed{\sigma_2 = 4} \rightarrow (3 \text{ marks})$$

$$\boxed{C.V._1 = \frac{\sigma_1}{\bar{x}_1} = \frac{3}{15} = 0.2}$$

$$\boxed{C.V._2 = \frac{\sigma_2}{\bar{x}_2} = \frac{4}{16} = 0.25}$$

$C.V._1 < C.V._2$, group 1 is more consistent.

(1 mark)

Q3

X - number of true-false question answered correctly

Y - number of mcq answered correctly

$X \sim \text{Bin}(4, 0.5)$

$Y \sim \text{Bin}(6, 0.25)$

$$P(X+Y=2) = P(X=0, Y=2) + P(X=1, Y=1) + P(X=2, Y=0)$$

$$= 0.01854 + 0.08899 + 0.06674 = 0.17427$$

$$P(X=0, Y=2) = {}^4C_0 \cdot 0.5^4 \times {}^6C_2 (0.25)^2 (0.75)^4 = 0.01853943$$

$$P(X=1, Y=1) = {}^4C_1 (0.5)^1 (0.5)^3 \times {}^6C_1 (0.25)^1 (0.75)^5 = 0.08898926$$

$$P(X=2, Y=0) = {}^4C_2 (0.5)^2 (0.5)^2 \times {}^6C_0 (0.25)^0 (0.75)^6 = 0.06674194$$

(A)

~~Q3~~ Q4
 Let $X = \text{number of}$

$$f(x, y) = \frac{1}{2} e^{-y} \quad \begin{matrix} y > 0 \\ -y < x < y \end{matrix}$$

a) if $x < 0$ $f_X(x) = \int_{-x}^{\infty} \frac{1}{2} e^{-y} dy$ (1.5)

$$= \frac{1}{2} \frac{e^{-y}}{-1}$$

if $x > 0$ $f_X(x) = \int_x^{\infty} \frac{1}{2} e^{-y} dy$
 $= \frac{1}{2} e^{-x}, x > 0$ (1.5)

$$f_X(x) = \frac{1}{2} e^{-|x|}, -\infty < x < \infty$$

(b) $f_Y(y) = \int_{-y}^y \frac{1}{2} e^{-y} dx = y e^{-y}, 0 < y < \infty$ (2)

(c) $\left. \begin{matrix} E(X) = 0 \\ E(Y) = 2 \end{matrix} \right\}$ (1)

$$E(XY) = \int_0^{\infty} \int_{-y}^y xy \frac{1}{2} e^{-y} dx dy$$

$$= \int_0^{\infty} \frac{1}{2} e^{-y} \left(\int_{-y}^y x dx \right) dy = 0$$

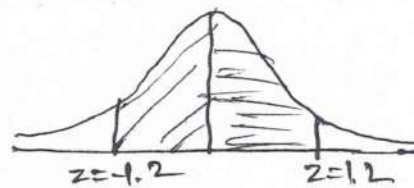
(2)

$$\text{Cov}(XY) = E(XY) - E(X)E(Y) = 0$$

Solution of Q.5

B/b) Given, $X = 0.496$, $\mu = 0.508$, $\phi(z) = 0.3849 = P(0 < z < 1.2)$
 $\mu = 0.502$, $\sigma = 0.005$, $n = 200$

Standard Normal variate $z = \frac{X - \mu}{\sigma}$



$$z_1 = 0.496 \text{ in standard units} = \frac{0.496 - 0.502}{0.005} = -1.2 = z$$

$$z_2 = 0.508 \text{ in standard units} = \frac{0.508 - 0.502}{0.005} = 1.2$$

— Thus the proportion of nondefective washers

= (area under normal curve b/w $z = -1.2$ and $z = 1.2$)

= (twice the area b/w $z = 0$ and $z = 1.2$)

$$= 2(0.3849) = 0.7698 = 77\%$$

— Therefore the percentage of defective washers is

$$= 100 - \text{nondefective washers}$$

$$= 100 - 76.98$$

$$= 23.02 = 23\%$$

c/e) Given — two independent random variables X and Y are both normally distributed, thus

Since $X \sim N(1, 9)$ and $Y \sim N(2, 16)$ are independent

Thus, $z = (X - Y) \sim N\{(\mu_1 - \mu_2), (\sigma_1^2 + \sigma_2^2)\} \sim N(-1, 25)$

if X follows the Normal distⁿ with parameters μ, σ^2 then pdf of X is given by,

$$p(x) = \begin{cases} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} & ; -\infty < x < \infty \\ 0 & ; \text{otherwise} \end{cases}$$

For density of X & Y — (1)

pdf of $Z = (X - Y)$ is given by $\{(X - Y) - (\mu_1 - \mu_2)\}$

$$p(z) = \begin{cases} \frac{1}{(\sigma_1 + \sigma_2) \sqrt{2\pi}} e^{-\frac{(X - Y) - (\mu_1 - \mu_2)}{(\sigma_1 + \sigma_2)} t^2} & ; -\infty < Z < \infty \\ 0 & ; \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{Z+1}{5} \right)^2} & ; -\infty < Z < \infty \end{cases}$$

M.G.F. of Normal distⁿ, if $X \sim N(\mu, \sigma^2)$, is given by

$$M_X(t) = e^{\mu t + \frac{t^2 \sigma^2}{2}} = \exp \left\{ \mu t + \frac{1}{2} \sigma^2 t^2 \right\}$$

Thus M.G.F. of $Z \sim N(\mu_1 - \mu_2, [\sigma_1^2 + \sigma_2^2])$ $Z \sim N(-1, 25)$

$$M_Z(t) = e^{-t + \frac{25}{2} t^2} = \exp \left\{ -t + \frac{25}{2} t^2 \right\}$$

A) 1

Given average of α -particles off is 3.2, we can see that to a very close approximation, the number of α -particles given off will be a Poisson random variable with parameter $\lambda = 3.2$.

The probability mass function of Poisson distⁿ is given by

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x=0, 1, \dots, \infty$$

$$P \{ \text{no more than 2 } \alpha\text{-particles will appear} \} = P(X \leq 2)$$

$$= P(X=0) + P(X=1) + P(X=2)$$

$$= e^{-3.2} + 3.2 e^{-3.2} + \frac{(3.2)^2}{2} e^{-3.2}$$

$$= e^{-3.2} \left(1 + 3.2 + \frac{(3.2)^2}{2} \right)$$

$$\approx 0.3799$$

Answer 6

Let, $U = X_1 + X_2$ and $V = X_2 + X_3$

As, X_1, X_2, X_3 are uncorrelated, $\text{Cov}(X_1, X_2) = 0, \text{Cov}(X_2, X_3) = 0, \text{Cov}(X_3, X_1) = 0$

Hence, $\text{Var}(U) = \text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$

$$= \sigma_1^2 + \sigma_2^2 \quad \text{--- (1)}$$

Again, $\text{Var}(V) = \text{Var}(X_2 + X_3) = \text{Var}(X_2) + \text{Var}(X_3) + 2\text{Cov}(X_2, X_3)$

$$= \sigma_2^2 + \sigma_3^2 \quad \text{--- (1)}$$

$$\text{Cov}(U, V) = E[(U - \text{mean}(U))(V - \text{mean}(V))]$$

$$= E[(X_1 + X_2 - \text{mean}(X_1 + X_2))(X_2 + X_3 - \text{mean}(X_2 + X_3))]$$

$$= E[(X_1 - \mu_{X_1}) + (X_2 - \mu_{X_2})][(X_2 - \mu_{X_2}) + (X_3 - \mu_{X_3})]$$

$$\text{Cov}(U, V) = \text{Cov}(X_1, X_2) + \text{Cov}(X_2, X_2) + \text{Cov}(X_1, X_3) + \text{Cov}(X_2, X_3)$$

$$= \sigma_2^2 \quad \text{--- (2)}$$

All computation must be shown in copy otherwise marks will be deducted.

$$1+1+2+1=5$$

$$\text{Hence, correlation coefficient} = \frac{\text{Cov}(U, V)}{\text{s.d}(U) \times \text{s.d}(V)} = \frac{\sigma_2^2}{\sqrt{\sigma_1^2 + \sigma_2^2} \times \sqrt{\sigma_2^2 + \sigma_3^2}} \quad \text{--- (1)}$$

where s.d stand for standard deviation of U and V respectively.

Answer 7.a.

Regression equation of X_1 on X_2 and X_3 is given by

$$(X_1 - \bar{X}_1) \frac{\omega_{11}}{\sigma_1} + (X_2 - \bar{X}_2) \frac{\omega_{12}}{\sigma_2} + (X_3 - \bar{X}_3) \frac{\omega_{13}}{\sigma_3} = 0 \quad \text{--- (1)}$$

$$\text{where } \omega = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}, \omega_{11} = \begin{vmatrix} 1 & r_{23} \\ r_{32} & 1 \end{vmatrix} = 1 - r_{23}^2 = 0.686$$

$$\text{similarly, } \omega_{12} = - \begin{vmatrix} r_{21} & r_{23} \\ r_{31} & 1 \end{vmatrix} = r_{13} \cdot r_{23} - r_{21} = -0.576, \text{ and } \omega_{13} = -0.048 \quad \text{--- (1)}$$

Required equation of plane of regression of X_1 on X_2 and X_3 is given by

$$(X_1 - 28.02) \frac{0.686}{4.42} + (X_2 - 4.91) \frac{-0.576}{1.10} + (X_3 - 594) \frac{-0.048}{85} = 0 \quad \text{--- (1)}$$

$$0.155X_1 - 0.524X_2 - 0.000564X_3 = 1.4352 \quad \text{--- (1)}$$

$$X_1 - 3.38X_2 - 0.00364X_3 = 9.2594$$

All computation must be shown in copy.

Answer 7. b. We have to find Regression line of x on y .

Consider the line $x = a + by$ (1)

Let, $x_i = a + by_i$

$$\Rightarrow \sum_{i=1}^n x_i = \sum_{i=1}^n a + b \sum_{i=1}^n y_i$$

$$\Rightarrow \bar{x} = a + b\bar{y} \text{(2)}$$

Also, $x_i y_i = a y_i + b y_i^2$

$$\Rightarrow \sum x_i y_i = a \sum y_i + b \sum y_i^2$$

$$\Rightarrow \frac{1}{n} \sum x_i y_i = a \bar{y} + \frac{b}{n} \sum y_i^2 \text{(3)}$$

From equation (2) we see that this line of regression of x on y passes through $(\bar{x}, \bar{y})$. — (1)

$$\text{Now, } \text{cov}(x, y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

$$\Rightarrow \frac{1}{n} \sum x_i y_i = \text{cov}(x, y) + \bar{x} \bar{y} \text{(4)}$$

$$\text{Again, } \sigma_y^2 = \frac{1}{n} \sum y_i^2 - (\bar{y})^2$$

$$\Rightarrow \frac{1}{n} \sum y_i^2 = \sigma_y^2 + (\bar{y})^2 \text{(5)}$$

From (3) and (4),

$$\text{cov}(x, y) + \bar{x} \bar{y} = a \bar{y} + \frac{b}{n} \sum y_i^2 = a \bar{y} + b(\sigma_y^2 + \bar{y}^2) \text{(6)}$$

Multiplying (2) by $\bar{y}$ and subtracting from (6), we get,

$$\text{cov}(x, y) = b \sigma_y^2 \Rightarrow b = \frac{\text{cov}(x, y)}{\sigma_y^2} \text{ — (1)}$$

Since b is the slope of the equation (1) and it passes through $(\bar{x}, \bar{y})$, therefore, equation of regression line of x on y is,

$$x - \bar{x} = b(y - \bar{y}) = \frac{\text{cov}(x, y)}{\sigma_y^2} (y - \bar{y}) = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}), \text{ since } r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$\Rightarrow x - \bar{x} = b_{xy} (y - \bar{y}), \text{ where, } b_{xy} = r \frac{\sigma_x}{\sigma_y} \text{ — (1)}$$

$$1+1+1 = 3$$

All steps must be there in copy.